

Mike Gao

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EDUCATION

University of Waterloo

Bachelor of Computer Science

Waterloo, ON

Sep. 2022 – Present

- Expected Graduation: May 2027

TECHNICAL SKILLS

Languages: C/C++, Python, Kotlin, JavaScript, Typescript, Java, C#, HTML/CSS

Frameworks/Tools: React, Next.js, Node.js, Express.js, Django, Git, Bash, Flask, Vercel, Firebase, Matplotlib, NumPy, SciPy, PyTorch, Gradle, Figma, Jupyter, Colab

EXPERIENCE

DupliCALL Co.

AI Software Engineer

Remote

Feb. 2025 – May 2025

- Conducted R&D on generative AI models for customer service, including dialogue summarization, service quality evaluation, and automated FAQ generation from multi-source data.
- Prototyped an intelligent call center assistant tool using **Python** and **Django**, integrating AI-driven keyword detection, FAQ retrieval, and real-time response generation to support human agents.
- Researched and integrated Google Cloud AI technologies (**Dialogflow CX**, **Vertex AI**, **Agent Assist**) for speech transcription, conversation analysis, and agent performance evaluation.
- Developed and benchmarked audio processing systems for speaker diarization, and noise suppression using open-source ML frameworks such as **Pyannote** and **SpeechBrain**.

R&D Engineer

Jan. 2024 – Apr 2024

- Researched, locally deployed, and extensively tested **90+** dialogue summarization LLMs using **Huggingface** and other open-source libraries.
- Developed innovative concepts to handle company requirements such as hardware optimization, multilingual translation capabilities, and integration into company call transcribers.
- Streamlined LLM testing methods through development of an interactive **tkinter GUI interface**, improving testing efficiency by **60%**.

PROJECTS

Timestamp 🗓️ | Kotlin, Jetpack Compose, Gradle, Spring Boot, Firebase, Docker

Sep 2024 – Dec 2024

- Spearheaded a team of 4 software developers to develop a secure time-management mobile app that organizes events, sends timely reminders, and provides real-time user location updates.
- Ensured smooth connection of Frontend and Backend with MVVM architecture & activity-scoped ViewModels.
- Built an intuitive frontend UI using **Material3** design, including Mapview page that leverages Google Maps API.

Chess ♟️ | C++, Git, Flask

Jul. 2024 – Aug. 2024

- Developed C++ chess application using OOP principles, utilizing inheritance and polymorphism principles for player/piece type hierarchy.
- Utilized MVC and Observer design patterns for user interactions, enhancing reusability/maintainability of code.

HONOURS & AWARDS

Finalist - UofTHacks 🏆 🦊 | JS, TS, React, Node.js, Next.js, Vercel

Jan. 2025

- Created Phish-Eye Lens - a phishing puzzle web-app that leverages **Gemini API** to generate & simulate unique user data for password-cracking clues.
- Employed **blackbox** LLM-tuning methods to ensure the model protects the puzzle solution regardless of user interference (through live LLM chat roleplay) but maintains intuitive and creative scenarios.

Best Use of APIs - GeeseHacks 🏆 🦊 | React, Node.js, Express.js, VertexAI

Jan. 2025

- Developed an AI-powered employee onboarding assistant that guides new hires through their workspace setup, offering live audio instructions and monitoring their progress in real time through their display.
- Adapted industry-leading LLM prompting methods and **RAG** techniques fine-tuned for live audio guidance, increasing output accuracy by **80%**.
- Integrated **Google Cloud APIs** (**TTS**, **STT**, **Vertex AI**) and a custom client to enable real-time audio/video interaction and seamless progress monitoring.
- Designed a **RESTful API** with **Express.js** for unique server endpoints, supporting Frontend file uploads and audio/video stream management.